

KYLER FUNG

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Education

San Diego State University

B.S. in Statistics (Emphasis in Actuarial Science) and B.S. in Accounting - Double Major

GPA: 3.70

Skills

- Programming: Python, R (RStudio, R Markdown), SAS, STATA, MATLAB, SQL, Excel
- Quantitative Methods: Stochastic Processes, Time Series Analysis, Regression Analysis, Markov Chains, RandomForest, Actuarial Risk Modeling, Financial Econometrics
- Accounting: Financial Statement Analysis, Taxation, Audit Procedures, Internal Controls

Research Experience

Undergraduate Researcher

Dr. Barbara Bailey

Aug 2025 - Feb 2026

- Developed R-based machine learning pipelines to analyze COPD patient data using unsupervised and supervised learning techniques.
- Performed missing-data imputation using median imputation followed by rImpute to improve data quality and model performance.
- Applied Partitioning Around Medoids (PAM) clustering and Random Forest to identify five distinct COPD patient subgroups.
- Validated cluster structure using supervised learning methods and prepared the framework for incorporating demographic variables into future analyses.

Undergraduate Researcher

Dr. Naveen Vaidya

Aug 2025 - Feb 2026

- Developed epidemiological models for Dengue Fever using systems of differential equations
- Extended deterministic ODE to stochastic settings to capture environmental and population uncertainty
- Modeled temperature dynamics as an Ornstein-Uhlenbeck stochastic process
- Formulated stochastic differential equation (SDE) models for mosquito population dynamics driven by temperature variability
- Implemented Euler-Maruyama numerical schemes and compared stochastic trajectories with deterministic ODE
- Used stochastic mosquito population forecasts to project infection dynamics over time

ARC Summer Researcher

Dr. Tomoyuki Ichiba

June 2025 - Aug 2025

- Computed asset pricing errors using the Fama-French Five-Factor Model
- Retrieved and processed portfolio return and factor data from the Kenneth French Data Library
- Compared path-signature features against GARCH volatility models for financial time-series prediction
- Designed hybrid models combining signature methods and GARCH outputs to assess predictive performance
- Evaluated trade-offs between signature depth, noise sensitivity, and computational complexity

Leadership Experience

Society of Statistics and Actuaries (SSA)

President

Aug 2025 - May 2026

- Founded the Society of Statistics and Actuaries with faculty mentorship
- Led journal-style discussions on applied statistics and actuarial topics
- Organized R-based replication projects of published academic studies
- Coordinated guest lectures from industry professionals and academic researchers

Poster Presentations

- "Optimal Investing Strategy using Asset Pricing Models," ARC Summer Research Poster, 2025
- "Do Stochastic Models Improve Dengue Vector Dynamics?" SDSU Student Symposium, 2025